# Correlators of boosted resonances

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## 1 Motivation

Measuring EECs of vector bosons is highly appealing. Potential applications include:

- Search for boosted narrow resonances decaying hadronically (as a bump hunt in the EEC spectrum). (Are we sensitive to Higgs??).
- Calibration/proof-of-principle for EEC top mass measurement. The actual top mass measurement is a separate work, this is just a first demonstration of the technique.
- Measurement of transverse/longitudinal polarization fraction of high-pT Z and W
- Measurement of the full production spin-density matrix for high-pT Z and W bosons

## 2 Background

The two-point energy correlator (EEC) of a boosted, hadronically decaying boson looks very different from that of a QCD jet, whose collinear spectrum is smooth and approximately scale invariant [1]. A boson of boost $\gamma = E_V/m_V$ instead produces a decay peak at $z' \simeq 1/\gamma^2$, with $z' = (1 - \cos\theta')/2$, which at central rapidity corresponds to $\Delta R \simeq 2m_V/p_T$ [1]. The width of the peak is set mainly by QCD radiation from the quarks rather than by $\Gamma_V/m_V$, and well below the peak the spectrum follows that of a quark jet [2].

### 2.1 Polarization and production

In the narrow-width approximation, the correlator factorizes into a production density matrix and a decay matrix [1, 2]. The opening-angle spectrum depends on production only through the longitudinal fraction $\rho_{00}$ [1]. Using NNLO production for $pp \to V + X$ and decay correlators at NNLO with NNLL collinear and N$^4$LL back-to-back resummation, Gao et al. find that the peak height at $\gamma = 4$ differs by about 70% between purely longitudinal and purely transverse bosons. For $p_T > 100$ GeV at $y_V = 0$, the polarized spectrum is suppressed by up to about 10% across the peak relative to an unpolarized treatment, because high-$p_T$ production is dominantly transverse [1]. The off-diagonal entries require measuring the detector orientation relative to the production plane [1]: $|\Delta h| = 2$ interference appears in the azimuthal angle, while $|\Delta h| = 1$ interference requires the azimuth and the angular separation jointly [2]. One-point energy and charge correlators together determine all eight real parameters of the density matrix [1]. Two experimental cautions come from Ref. [2]: boosts along the LHC beam axis modify the azimuthal distributions, an effect expected to be mild but not yet assessed, and in a lepton-collider study $D_2$ selections preferentially reject transverse bosons, homogenizing the longitudinal and transverse one-point spectra.

Two neighboring results extend the toolkit. The one-point correlator of a vector current is fixed by a single parameter $a_\mathcal{E}$, bounded by unitarity to $-1/2 \le a_\mathcal{E} \le 1$; its N$^3$LO value

at $q^2 = m_Z^2$ is $-0.42$, in good agreement with values extracted indirectly from LEP data [3]. Flavor flux correlators generalize the program to flavor quantum numbers, with hadronization corrections to the $b$ and $c$ cases suppressed by $\Lambda_{\mathrm{QCD}}/m_Q$ [4].

### 2.2 Searches

The same peak motivates resonance searches. For a narrow resonance coupled linearly to a QCD operator, the peak position encodes the mass and the spectral shape encodes the spin and polarization, within a space of shapes bounded by unitarity and energy positivity [5]. A projection for a light hadrophilic $Z'$ in $pp \to Z'\gamma$, with statistical uncertainties only, finds that the inclusive EEC reaches sensitivity comparable to a recast of a CMS soft-drop mass search [5]. A separate recast of the CMS dijet correlator measurement also gives competitive limits, although its signal is at particle level and detector effects may degrade the reach [5]. In a Monte Carlo study of jets with $p_T > 1$ TeV recoiling against $Z \to \ell\ell$, hadronic $W/Z$ decays appear as a local excess of about 7% even though they make up only about 1% of the jets, with no reconstruction or tagging; that selection is better suited to the HL-LHC than to existing datasets [1]. The peak position also calibrates the angular scale, making $m_V$ a standard candle inside jets, including the polarized $W$ inside boosted top jets [1].

## 3 Questions and Goals

In the short term, we want to understand how we can use these ideas to do a CMS measurement. A few important lines of questioning:

**"Untagged" resonance search**

- What sensitivity might we expect in an "untagged"-style search for a narrow resonance in Run 3 data?
- Which dataset(s) would we want to use?
  - Z+jet (tagged by Z$\to \ell\ell$)? Potential advantage: polarization of the other Z from leptonic decay?
  - W+jet (tagged by W$\to \ell\nu$)? Potential advantage: polarization of the other W from leptonic decay?
  - Photon+jet (tagged by photon)? Potential advantage: larger cross-section. Photon cannot be longitudinally polarized, I assume this has some implications for the polarization of the Z?
  - Dijet (with exactly two jets and momentum balance)? Potential advantage: large cross-section, but also large QCD background.
  - QCD multijet (without momentum balance)? Potential advantage: largest cross-section, but also largest QCD background.
  - Something else?
- What do various SM processes look like?
  - Can we separate W and Z at all?
  - Can we see the Higgs? If not, can we set limits? Would they be competitive?
  - What does the top look like? Do we learn anything interesting?

- Do we get any extra sensitivity from observables beyond two-point correlator? E.g., E3C, E1C, etc. Including non-standard formulations thereof? The paper mentions $E^2E^2$ for example. Or with non-standard angles as in the one-point from the paper...
- Would we get a boost in sensitivity from some additional cuts?
- Is heavy flavor interesting? Can we see heavy mesons?
- Is the background really as clean as we hope it is?
- How bad is the jet energy uncertainty? Is that the dominant systematic? What other systematics could be important?
- Would our limits actually be competitive?
- What jet radius to use? I assume AK8
- Use tracks or PF candidates?
- Useful to discriminate signals by spin/polarization? Or just an unpolarized bump hunt? Separate limits for vector vs scalar? What would an axial-vector signal look like? Or a pseudoscalar?

**Polarization measurements and spin density matrix**

- How good of a job do we need to do at tagging? Can we do it untagged? Is there any way to do good tagging that doesn't badly sculpt the distributions?
- Is there a useful shape difference between longitudinal and transverse bosons? Or is it dominantly the peak height?
- Are the one-point correlators actually practical? What does the background do to them?
- Would the three-point correlator be useful/interesting? Or some related things, maybe non-standard ones...
- Is there an interesting bell inequality story here? Can we see interference effects?
- Again, which dataset(s) would we want to use?
- Charge correlators? Are they useful? How badly does hadronization etc. ruin the picture?
- How large are the spin analyzer coefficients?
- What is the other literature? Are we competitive?
- What jet radius to use? I assume AK8
- Use tracks or PF candidates?
- What is the best approach for getting the full matrix? Minimal set of most useful correlators? Compare one-point to two-point to three-point?
- Link to EFT?

# 4 Tasks

We will refine the procedure as we learn, so each step below ends with a check-in to review the results and revise the later steps. Quantitative metrics for comparing channels, selections, and observables will be defined at these check-ins, once the procedure is better understood.

1. **Literature.** Read more literature to get a sense of what has already been done, and fix the mass and $p_T$ ranges of interest.
   - What are the current limits on narrow resonances in the mass range of interest? Start from the references in Ref. [5].
   - What are the current uncertainties on the W and Z longitudinal fractions, as a function of $p_T$? Start from the references in Ref. [1].
   - What are the current uncertainties on the full spin density matrix, as a function of $p_T$?
   - *Check-in.*
2. **Reproduce and explore.** Set up hadron-level correlator code, starting with only the Z($\ell\ell$) channel, and scale out from there.
   - Required samples: Z($\ell\ell$)+jet, WZ, and ZZ, where the diboson samples have Z$\to \ell\ell$ and a hadronic W or Z. All of these are needed, because the published result is built from their sum.
   - Two-point correlator: reproduce the Z($\ell\ell$)+jet result in Fig. 8 of Ref. [1].
   - One-point correlator: explore it ourselves in the same samples. We are not targeting an exact reproduction of published one-point results, which use different samples [2].
   - *Check-in.*
3. **Samples and data formats.**
   - Survey which of the samples in Table 1 are already available centrally, and use them when possible.
   - We will probably need to skim a custom NANO format from MiniAOD to get at the jet constituents. Check whether everything we need is in MiniAOD, or if we need full AOD.
   - *Check-in.*
4. **Polarization feasibility.** Using longitudinal and transverse samples, determine how sensitive the correlators are to the longitudinal fraction.
   - Explore and experiment with how to track boson polarization in the generation, which is not yet clear.
   - Check empirically how the choice of jet radius affects longitudinal and transverse bosons.
   - *Check-in:* decide whether the SM polarization measurement remains a goal.
5. **Search exploration.** Add the remaining samples a few at a time, and experiment with different channels, jet radii, selections, cuts, and observables to see what is most promising, using a jet-mass analysis as a baseline.
   - *Check-in.*

6. **Detector-level studies.** Move to the central CMS detector simulation, compare parton, hadron, and detector level for everything, and add systematic uncertainties, a background estimation method, and a statistical model that accounts for correlations between bins.
   - *Check-in.*
7. **Proposal.** Based on these studies, set up a minimal proposal for a measurement and/or search, including a sensitivity estimate. Rediscovering the W and Z is the natural first milestone.

**Samples** Table 1 lists the samples we expect to need. Take care in sample generation to keep track of the polarization state of every boson we simulate.

Table 1: Planned simulated samples.

| Sample | Decays and variants | Notes |
|---|---|---|
| $Z'$ resonances | Small number of mass points to start | |
| Scalar resonances | Small number of mass points to start | |
| Higgs | Various decay channels | Can we see decay to light quarks? |
| Z+jet | $Z\to \ell\ell$ and $Z\to q\bar{q}$, separately | $Z\to \ell\ell$ required for step 2 |
| W+jet | $W\to \ell\nu$ and $W\to q\bar{q}'$, separately | |
| V$\gamma$ | Leptonic and hadronic V decays, separately | |
| Photon+jet | | |
| Diboson | All permutations of leptonic and hadronic decays | WZ and ZZ with $Z\to \ell\ell$ and a hadronic W or Z required for step 2 |
| Multiboson | All permutations of leptonic and hadronic decays | |
| QCD multijet | | |
| Single-top | All channels | |
| ttbar | | |

# 5 Resources

This is a first draft, and the choices marked as open will be settled at the check-ins above.

### Data and simulation

- CMS Run 3 data, at MiniAOD level, or AOD if the check in step 3 shows that we need it.
- Centrally produced CMS simulation, wherever the samples in Table 1 are available.
- Privately generated samples for anything else, likely including the resonance signals and the polarized bosons. These start at LO and at hadron level, with no detector simulation. We can graduate to aMC@NLO, POWHEG, or similar once we have a handle on things.
- Comparisons of parton, hadron, and detector level for everything, eventually using the central CMS detector simulation.

**Software**

- Generation: MadGraph5_aMC@NLO at LO and Pythia 8, the generators used in the published studies [1, 5]. How to track boson polarization is open, and we will need to explore and experiment.
- Jet clustering and correlators: FastJet and our own correlator code.
- CMS software: CMSSW for the custom NANO skim from MiniAOD or AOD.
- Analysis: a Python columnar stack (uproot, awkward-array, hist).
- Open: statistical inference, which will take some iteration.

**Computing**

- Batch computing and storage for sample generation and for the custom NANO skims. The site will likely be FNAL LPC, but we will evaluate the options available. The required capacity will be estimated once the sample survey in step 3 shows how much we need to generate.
- Enough events in the high-$p_T$ tails, which will likely require generating the V+jet and QCD samples in $p_T$ slices.
- CMS grid credentials and access to central datasets.

**Theory inputs**

- Polarized two-point predictions: for now, compare against the plots in Ref. [1].
- The unitarity-bounded classification of resonance signal shapes from Ref. [5].

**People** Simon Rothman, for now.

## References


[1] Anjie Gao, Kyle Lee, and Xiaoyuan Zhang. *Precision Studies of Boosted Polarized Vector Bosons with Energy Correlators*. 2026. arXiv: 2610.38688 [hep-ph].

[2] Lorenzo Ricci and Marc Riembau. "Energy Correlators of Hadronically Decaying Electroweak Bosons". In: *Physical Review D* 106.11 (2022), p. 114010. DOI: 10.1103/PhysRevD.106.114010. arXiv: 2207.03511 [hep-ph].

[3] Marc Riembau and Minho Son. "Flow between Extremal One-Point Energy Correlators in QCD". In: *Physical Review Letters* 136.10 (2026), p. 101901. DOI: 10.1103/lc5m-kwgv. arXiv: 2509.02669 [hep-ph].

[4] Kyle Lee and Ian Moult. *Flavor Flux Correlators, Forward-Backward Asymmetries, and Anomalies*. 2026. arXiv: 2609.13366 [hep-ph].

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